NIKITHA LOHIA

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	10.00
Narayana Junior College	Class XII (Board of Intermediate Education, A.P.)	2014	92.20%
Kendriya Vidyalaya, Shivrampally, Hyderabad	Class X (CBSE)	2012	8.80%

SCHOLASTIC ACHIEVEMENTS

1. Awarded the prestigious Prime Minister's Research Fellowship (PMRF).
2. Received the Institute Gold Medal for Academic Excellence during the M.Tech. program.
3. Achieved a perfect 10.0 CGPA in Ph.D. coursework.
4. Second best award for 3 minutes presentation @Impact180 on Chemical Engineering Department Day, IITH 2025.

PROJECTS

Ph.D. Projects:

1. Explored the microdynamics of dilute colloidal gels with the objective of actively tuning their elastic modulus by probing internal gel dynamics via dynamic light scattering under systematically varied salt concentrations, thereby establishing structure-property relationships in colloidal networks.
2. Developed and tested a physics-based framework linking intermolecular forces, fluid-mechanical flow, and interfacial phenomena to the emergence of intrinsic fractal networks in evaporating sessile droplets, using high-resolution optical microscopy to correlate drying dynamics with final deposit morphology.
3. Formulated high-viscosity polymeric and particle-loaded inks and performed comprehensive linear and non-linear rheological characterization to construct a rheology-guided printability regime map, enabling rational design of soft complex inks for extrusion-based additive manufacturing.

M.Tech Project:

Conducted advanced Fortran-based numerical simulations of subcooled flow boiling at low and high pressures using a four-equation drift-flux model, quantifying the influence of operating pressure on interfacial heat transfer mechanisms and bubble departure diameter.

B.Tech Projects:

1. Investigated the influence of welding process parameters on the mechanical properties of dissimilar metal welds between AISI 316 and AISI 321 stainless steels, with emphasis on microstructural evolution and resulting performance characteristics.
2. Examined the flour milling process from a process-engineering perspective, identifying key operational bottlenecks and proposing practical measures to mitigate difficulties and improve throughput and reliability.

Industrial Project Training:

Completed structured project training at Bharat Heavy Electricals Limited (June 2018), gaining exposure to the full manufacturing workflow of steam turbines, including component fabrication, quality assurance, and assembly-line practices in a heavy engineering environment.

PUBLICATIONS

1. **Nikitha Lohia**, Mahesh Ganesan and Alan Ranjit Jacob; Internal dynamics of colloidal gel as a function of salt concentration, Manuscript in progress.
2. **Nikitha Lohia**, Mahesh Ganesan and Alan Ranjit Jacob; Gel-like network due to self- assembly in an evaporating droplet, Manuscript in progress.
3. Akshay Tikoo, **Nikitha Lohia**, Sri Surya Charan Kondeti, Praveen Meduri; Mechanistic insights of enhanced

photocatalytic HO production induced by Z- scheme heterojunction of copper bismuth oxide and molybdenum sulfide; Journal of Material Chemistry A; 2023, DOI: https://doi.org/10.1039/D3TA02875E

CONFERENCES

1. Nikitha Lohia, Mahesh Ganesan and Alan Ranjit Jacob; Bridging rheology and 3D printing of soft complex Inks, P038, Rheology and Nonequilibrium Phenomena; 10th International Soft Matter Conference, May 25-29, 2026, BITS Pilani KK Birla Campus, Goa.
2. Nikitha Lohia, Mahesh Ganesan and Alan Ranjit Jacob; Understanding complex fluids: From dilute to concentrated regime, PMRF Annual Symposium, 2026, IIT Guwahati.
3. Nikitha Lohia, Mahesh Ganesan and Alan Ranjit Jacob; Emergence of 2D Fractal Gel Networks in Evaporating Colloidal Systems, 106, Colloids, Gels, Emulsions, and Foams; COMPFLU, December 15 - 17, 2025, IISc Bangalore.
4. Nikitha Lohia, Mahesh Ganesan and Alan Ranjit Jacob; Microdynamic Insights into Gelation in Dilute Colloidal Systems, 332e; AIChE Annual Meeting, November 2 - 6, 2025, Boston, Massachusetts, USA.
5. Nikitha Lohia, Mahesh Ganesan and Alan Ranjit Jacob; Fractal-like Patterns in Drying Droplets of Colloidal Dispersions, 475i; AIChE Annual Meeting, November 2 - 6, 2025, Boston, Massachusetts, USA.
6. Nikitha Lohia, Mahesh Ganesan and Alan Ranjit Jacob; Exploring Apparent Universalities in Dilute Colloidal Gels, GN04, Gels & Networks; COMPFLU, December 16 - 18, 2024, IIT Hyderabad.

INTERNSHIPS

1. Currently pursuing a PhD Accelerator Internship at L'Oréal Research & Innovation, Bangalore, conducting rheo-tribological analysis of skin and hair care products and handling SEM-EDX and HPLC instrumentation.
2. Completed a six-month internship at Saboo RKS Motors, an authorized Maruti Suzuki dealership in Hyderabad, gaining hands-on knowledge of four-wheeler components and performing basic vehicle servicing.

SKILLS

1. Extrusionbased 3D Printing: Formulation of particleloaded and polymeric inks with tailored rheological properties for direct ink writing.
2. WetLab Synthesis: Particle synthesis via emulsion polymerization and thermomechanical methods; hydrothermal synthesis of catalysts.
3. Advanced Characterization: Dynamic Light Scattering (DLS), rotational rheometry, and optical microscopy for microstructural and flow behavior analysis.
4. Software & Design: AutoCAD and Cura for 3D printing workflow; MATLAB and FORTRAN for numerical simulations and data analysis.

POSITIONS OF RESPONSIBILITY

1. Managed optical microscopy facilities for the Soft Matter Group, IITH (2023-present), including instrument operation, basic maintenance, scheduling, and training of students and researchers.
2. Oversaw safe operation of the thermomechanical hot press in the Soft Matter Group (2023-present), providing technical support and guidance to internal and external users.
3. Served as Rheometer 302e trainer for SATHICISCoM, IITH (2025-present), conducting handson training sessions, assisting users in experimental design, and supporting data acquisition and interpretation.
4. Acted as chemical inventory manager for the Soft Matter Group (2024), responsible for procurement, receipt, storage, labeling, and tracking of chemicals to ensure laboratory safety and regulatory compliance.

RELEVANT COURSES

1. Attended the ISMC Mini-School on the theme Microscopy & Imaging, BITS Pilani, Goa, May 22-25, 2026.
2. Intensive online training on COMSOL Multiphysics in August 2023.
3. Workshop on Optical Microscopy organized by IITH- Nikon CoE and SATHI- CISCoM, March 2025.
4. COMPFLU 2024 pre-conference workshop on Physics and Chemistry of Food.

EXTRACURRICULAR

1. Volunteered as a Ph.D. buddy from Sept 2023 to April 2024 in Sunshine- The Counselling Cell of IITH.
2. NSS volunteer for two years during under graduation.
3. A2 German language certified, 2020.